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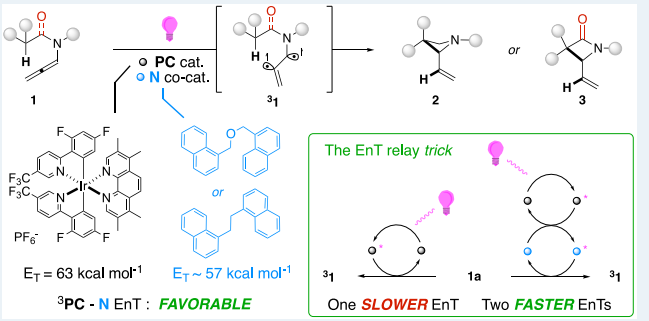
Synthesis of Vinyl Azetidines and β-Lactams from Allenamides via Energy-Transfer Relay

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ABSTRACT: Azetidines and β-lactams are useful and versatile building blocks with ample synthetic applications and pharmaceutical uses. However, few catalytic methods provide densely functionalized monocyclic derivatives, and none could easily afford both azetidines and β-lactams. Herein, we solved this issue with allenamides. The challenging substrates were activated by energy transfer combining a tailor-made Ir(III) complex and a binaphthyl cocatalyst, which acts as a triplet-state relay and stabilizes intermediates via dispersion interactions. A triplet intermediate with a vinyl-radical character gives the strained heterocycles upon 1,5 hydrogen atom transfer. Products have a congested all-carbon quaternary center and a useful allylamine unit and are prepared at room temperature with complete atom economy. The utility of the method is shown by further downstream derivatization.



KEYWORDS: *allenes, azetidines, energy transfer, hydrogen atom transfer, synthetic methods, synergistic catalysis, visible light*

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INTRODUCTION

Azetidines and β-lactams are key parts of bioactive molecules, worthy three-dimensional bricks for medicinal chemistry, and reagents for elaborate syntheses (Scheme 1a).[1] Few catalytic methods that give densely functionalized monocyclic products were reported. None could be readily switched to provide either azetidines or β-lactams, and the incorporation of an allylamine fragment, which is a vaunted structural feature,[2] is similarly underdeveloped.

Nucleophilic substitution has been widely used to create strained rings,[1,3] but the synthesis of functionalized precursors is an open issue. For instance, lengthy linear sequences were used to prepare the relatively simple α-vinyl azetidine,[3e,f] which is a highly versatile species (Scheme 1b).[4]

Elegant catalytic methods based on the release of ring strain have been presented.[5] *N*-protected azetidines can be accessed from β-iodo ynamides.[6] These approaches require, however, the presence of sacrificial groups. Higher atom economy has been achieved by a few energy-transfer (EnT)[7] methods. A strain-release approach uses [1.1.0]-azabicyclobutanes and an *N*-sulfonyl diarylimine.[8] An intermolecular [2 + 2] photocycloaddition combines a β-carboxy oxime with a styryl fragment.[9] Sensitized aryl-methacrylamides can deliver β-lactams via hydrogen atom transfer (HAT)[10] to an alkyl radical.[11]

So far, EnT strategies have been focused on the activation of conjugated π-bonds, and the required auxochromes, such as aryl rings and carbonyl groups (Scheme 1c, highlighted in orange), limit the chemical space that can be accessed by these methodologies.

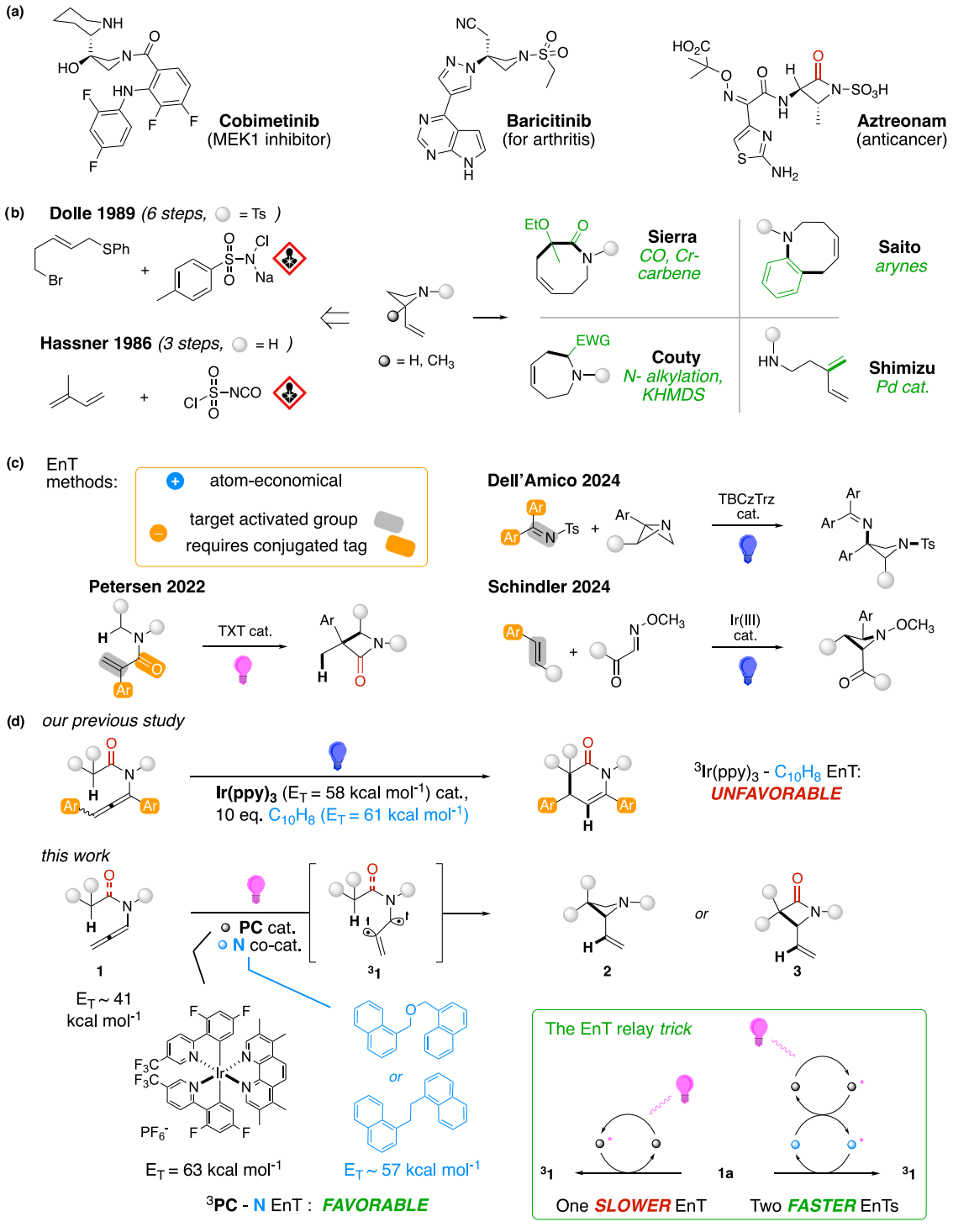
We targeted instead a *cumulated* π-system. A few reports show that EnT activation of an allene can form species with a vinyl-radical site.[12] We reported that bis-aryl allenamides can afford δ-lactams via 1,5-HAT in the presence of Ir(ppy)$_3$ and a molar excess of naphthalene.[13] The reaction was limited to conjugated trisubstituted allenes, and terminal allenamides did not react, likely because of their reduced conjugation (Scheme 1d). We wanted to find suitable conditions for the activation of unbiased allenyl substrates, aiming to create strained 4-membered heterocycles rather than δ-lactams, via sequential 1,5-HAT/radical recombination. This outline is challenging because the use of monosubstituted allenamides in EnT methods has few precedents,[12b,c] and none involves a HAT step. Nonetheless, the concept would bring advantages: the sequence would provide α-vinyl azetidines (or β-vinyl-β-lactams) and, at present, no general catalytic method has been reported[5,6,8,9,11] to synthesize these heterocycles, which present a synthetically useful allylamine unit;[2,4] additionally, in contrast to the conventional generation of vinyl radicals,[10,14]

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Scheme 1. (a) Examples of Relevant Azetidines and β-Lactams; (b) Syntheses and Applications of α-Vinyl Azetidine; (c) EnT Approaches Activating Conjugated π-Bonds; and (d) Our Previous Synthesis of δ-Lactams from Bis-aryl Allenamides vs Present EnT-Relay Activation of Unbiased Allenes



the EnT on an allene is atom-economical, and it would both trigger the desired HAT and provide the vinyl function to the product.

We report the first synthesis of monocyclic azetidines or β-lactams from allenes. The visible-light-promoted HAT sequence has broad functional group tolerance and delivers products with a challenging congested all-carbon quaternary center and an exocyclic vinyl group.

The method combines a tailored Ir(III) complex with high triplet energy (E_T) and a binaphthyl cocatalyst, which has two roles. It can stabilize intermediates via dispersion interactions; although dispersion interactions between closed-shell entities have an established impact in catalysis,[15] the exploitation of radical-π interactions has been vastly overlooked.[13,16] Additionally, the cocatalyst is a triplet-state relay that ensures a more efficient substrate activation. This feature is original and not previously observed in our studies because they involved

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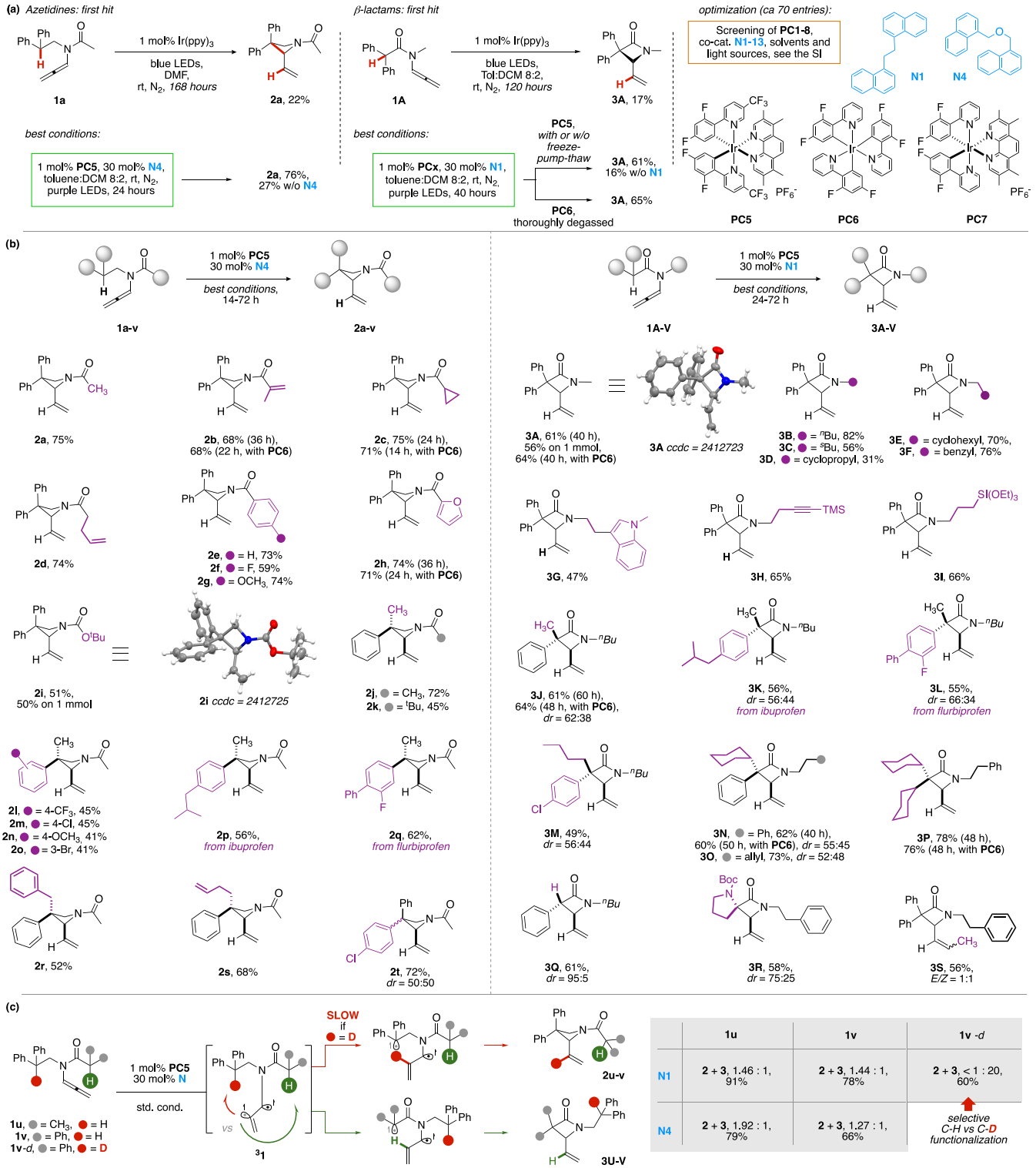


Figure 1. (a) Reaction discovery and summary of key parameters; conditions: 0.15 mmol **1**, 1 mol % PC, blue (λ_{max} = 458 nm) or purple LEDs (λ_{max} = 395 nm), ^{1}H NMR yields. (b) Scope of the synthesis of azetidines **2** (left) and β-lactams **3** (right), isolated yields. (c) **N** & primary kinetic isotope effects on the selectivity of the C(sp^3)−H functionalization.

commercial Ir(III) complexes whose E_T values were too low for the efficient sensitization of naphthyl derivatives.[12c,13] Furthermore, a triplet-state relay behavior has so far been restricted to reagents, rather than (co)catalysts that present an internal antenna, such as an acyl-naphthalene group.[17] Present results can thus lead to broad application by showing that the concept is not limited to specially designed substrates.

RESULTS AND DISCUSSION

In an initial experiment, a 0.1 M solution of **1a** and 1 mol % of Ir(ppy)$_3$ (**PC1**) were charged into a 5 mm NMR tube (Figure 1a, extended optimization in the SI). The mixture was degassed by freeze−pump−thaw and then irradiated with a household blue LED strip. A meager 22% yield of **2a** was

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observed upon 168 h of irradiation (Figure 1a). Both the product and the substrate partially decompose over time in the presence of light and the photosensitizer, hindering the final yield. We then began an optimization effort. A detailed presentation of this work, which involved the use of eight different Ir(III) photocatalysts (**PC1**−**8**) and 13 poly naphthalene cocatalysts (**N1**−**N13**), is presented in the SI together with a brief comment of the variation of each reaction parameter (see Table S1a−i). Overall, marginal improvements were achieved by varying the solvent, the catalytic charge, or the concentration of the mixture. The nature of the PC had a larger role, but none of the tested complexes alone was sufficient to achieve good results. However, a significantly faster conversion of **1a** was invariably observed in the presence of each of the 13 **N** derivatives (Table S1e,h), and this effect was general among different **PC**s (Table S1a,g).

The highest yield of **2a** was achieved by combining **N4** and **PC5**, under purple LED irradiation for 24 h. The complex has not been previously used in EnT methods despite its high E_T (63.0 kcal/mol)[18] and long-lived triplet state (τ = 3.8 μs in MeCN),[19a] which, together, suggested that it could be a useful tool. A faster conversion of **1a** (14 h), coupled with a slightly lower yield of **2a**, was observed combining **N4** with either homoleptic complex **PC6**[19b,c] or heteroleptic complex **PC7**,[19d] which has the same *C,N*-ligands of **PC6** and the same *N,N*-one of **PC5**. These results show that there is no linear correlation between a single photophysical property of the PC and the outcome of the reaction because **PC6** has the highest E_T of the **PC1**−**8** series (63.5 kcal/mol)[19c] and the shortest-lived triplet state (τ = 1.6 μs),[19b] while the opposite holds for **PC7** (58.0 kcal/mol and 21.7 μs for its E_T and τ, respectively).[19d] Finally, no traces of **2a** were formed without PC or a light source.

An analogous approach was used for the synthesis of β-lactam **3A**. The detailed presentation of the study is presented in SI, Table S2a−g. The optimization confirmed the key role of the **N** cocatalyst (Table S2c,f), in analogy to the observation made for the synthesis of azetidine **2a**. The best results were achieved by combining **N1** with either **PC5** or **PC6**. Interestingly, the use of the latter provided a slightly worse yield, but the outcome of the reaction proved insensitive to the presence of traces of dioxygen, which is a powerful quencher[20] of photoexcited Ir(III) complexes (for a possible rationale, vide infra). Indeed, the irradiation time required to consume **1A** and the yield of **3A** were identical (61% upon 40 h) if the reaction mixture was degassed by freeze−pump−thaw (3×), or not. A higher yield of **3A** could be achieved by employing **PC6** (65% upon 40 h), but the outcome became worse (59% upon 48 h) if the freeze−pump−thaw step was skipped (a similar trend was observed for **2a**, see Table S1i in the SI). It is worth noting that this thorough degassing procedure represents an intrinsic practical limitation for a synthetic method.

We next examined the scope of these reactions. Allenamides were prepared by sequential reductive amination, acylation, and isomerization. This route tapped into an ample reservoir of readily available aldehydes and acylating species. A large panel of either azetidines or β-lactams could thus become accessible by combining at will these different building blocks. The scope of these two reactions is presented in Figure 1b.[21]

The acetyl unit of **2a** could be replaced by a methacryloyl fragment with a minimal loss of efficiency (**2b**), and no polymerization took place. A strained cyclopropyl arm and terminal α-olefins were similarly tolerated (**2c**−**d**). The presence of an arylamide did not hamper the reaction (**2e**−**h**). Best results were achieved with electron-neutral and -rich (hetero)aryl rings, while a slightly lower yield was observed with electron-poor systems. The amido group could be replaced by a carbamate group (**2i**), which could be easily cleaved. The product was retrieved in a synthetically useful yield, and the reaction could be scaled up to 1 mmol. The variation at the β-carbon was then assessed. A tertiary C(sp^3)−H bearing one aryl substituent can undergo functionalization despite a higher BDE (**2j**).[14] The steric demand of the acyl group was more relevant in this case (**2k**). The presence of substituents on the aryl ring led to the formation of products in moderate yield (**2l**−**2o**). The method tolerates the presence of Ar−Br groups, which could be useful for further functionalization. Substrates derived from ibuprofen and flurbiprofen delivered the products **2p**−**q** with good competence. Pendant benzyl groups and alkenes were similarly tolerated (**2r**−**s**). A single diastereomer of products **2j**−**2s** was isolated. The quaternary carbon of the azetidine can present two different aryls (**2t**). Representative substrates **1b**, **c**, and **h** were also tested in the presence of **PC6** instead of **PC5**. As observed with **1a** (see SI, Table S1f), the irradiation time required to fully convert the reagent was reduced, but a slight erosion of the yield occurred. Substrates with a secondary C(sp^3)−H did not afford **2**, possibly owing to their higher BDE (limitations in the SI).[14]

We then switched to the synthesis of β-vinyl β-lactams (Figure 1b, right). Substrates with different *N*-arms afforded β-lactams in up to 82% yield (**3A**−**F**). An increased steric hindrance close to the nitrogen center reduced the efficiency of the reaction, which nonetheless tolerated a *sec*-butyl or cyclopropyl fragment (**3C**−**D**). Substrates with *N*-aryl arms performed poorly. On the contrary, β-lactams featuring an indole, an alkyne, or a silyloxy pendant were smoothly accessed (**3G**−**I**).

Substituted C(sp^3)−H groups could be functionalized, leading to the formation of the β-lactams in up to a good yield and with moderate diastereocontrol (**3J**-**O**). The series included bulky cyclohexyl fragments and poly-substituted aryls, such as those derived from two commercial drugs. The presence of an aryl substituent on the key C(sp^3)−H group is not necessary (**3P**). The functionalization of a secondary C(sp^3)−H group, which had a relatively higher BDE,[14] occurred diastereoselectively (**3Q**). The spirocycle[22] **3R** was prepared, showing that the method can be applied to amino acid derivatives. Finally, disubstituted allenes could be employed (**3S**). Representative substrates **1A**, **J**, **N**, and **P** were also tested in the presence of **PC6** instead of **PC5**. No clear trend emerged, and results were, on average, comparable, suggesting that the two complexes are almost equally competent for this reaction.

The reaction of **1u** afforded a 1.46:1 mixture of azetidine **2u** and β-lactam **3U** (Figure 1c). The **2**:**3** ratio was altered by varying **N** (up to 1.92:1, for a rationale, vide infra). The same trend was observed using compound **1v**. Both substrates had two competing C(sp^3)−H groups (green vs red −H in Figure 1c). Interestingly, the reaction of **1v**-*d* exclusively gave **3 V**-*d*. The *d*-labeled β-lactam was accessed by steering the selectivity of the C−H vs C−D functionalization, which is often challenging.[23] The result indicated that this step presents a large primary kinetic isotope effect.[24] This, in turn, is consistent with a 1,5-HAT, for which significant k_H/k_D (>7) were reported.[10,23b,24a]

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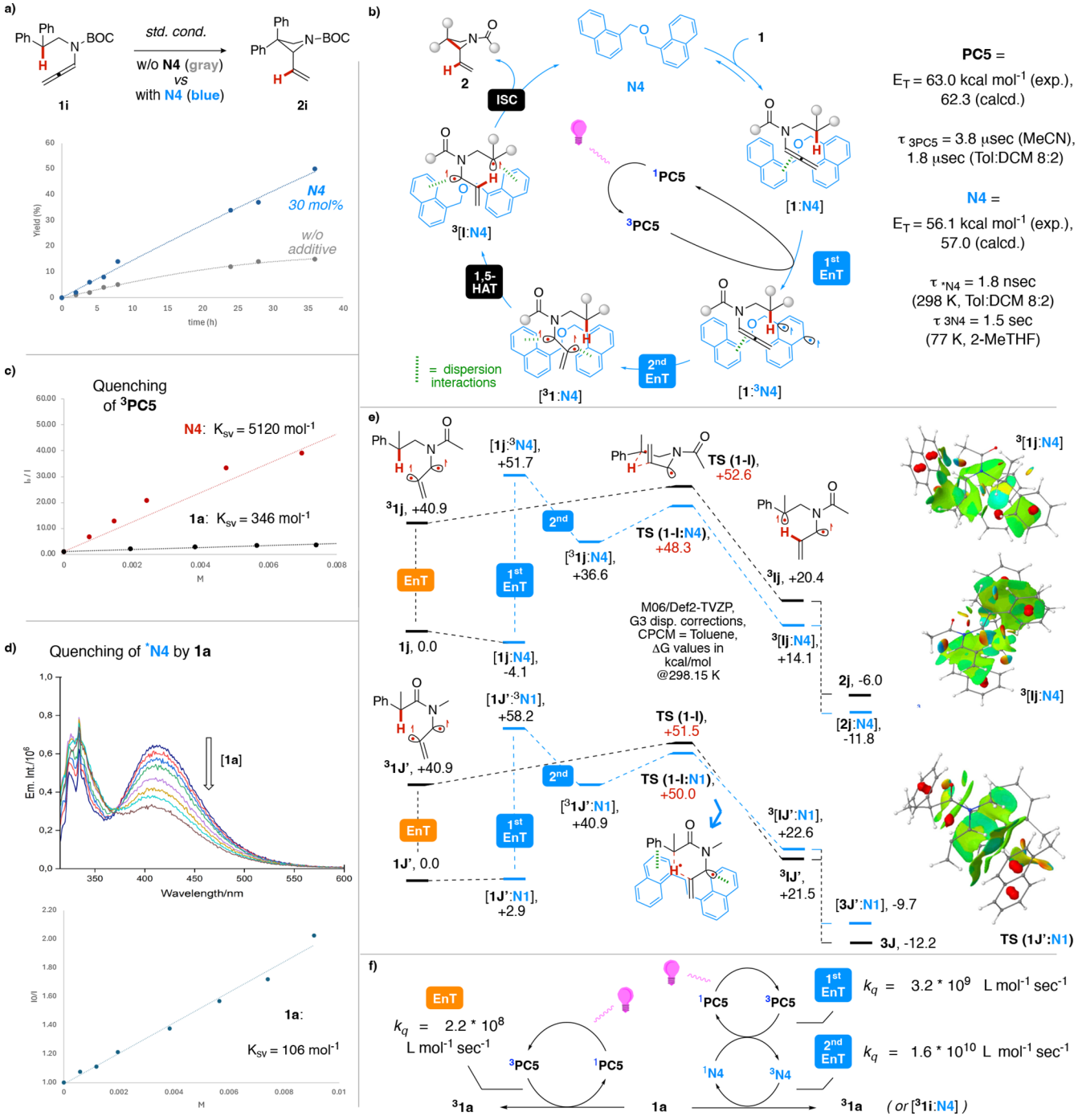


Figure 2. (a) kinetic study proving the cocatalytic effect; (b) proposed mechanistic rationale; (c) phosphorescence quenching studies on ^{3}**PC5**; (d) luminescence quenching studies on **N4**; (e) DFT modeling data with NCI plot of key intermediates, color code: green basins for dispersion interactions, red spheres for ring critical points, orange/blue lobes for destabilizing steric interactions; and (f) comparison of the kinetic quenching constants supporting the EnT relay scenario.

The catalytic role of **N4** was established by monitoring two parallel reactions through time by ^{1}H NMR (Figure 2a). The rate of formation of **2i** was sharply lower without **N4**.

This effect can be explained by the mechanism of formation of compound **2** presented in Figure 2b.

The **N4** and **1** can form the [**1:N4**] adduct thanks to stabilizing $\pi-\pi$ and C−H−π interactions.[25,26] Two sequential EnTs would follow. The first one is intermolecular and it involves the photoexcited ^{3}**PC5** and **N4**, giving [**1:**^{3}**N4**]. The second EnT is intramolecular, and it occurs between ^{3}**N4** and **1**, affording [^{3}**1:N4**] that has a delocalized allylic mono-occupied molecular orbital and a reactive vinyl-radical one. This intermediate can deliver [^{3}**I:N4**], which has a more stable tertiary benzylic radical site, via 1,5-HAT, in agreement with the results of the labeled substrate **1v-*d*** (Figure 1). The azetidine ring is established upon intersystem crossing via radical recombination, allowing **N4** to turn over. An analogous manifold affords β-lactams **3**, as calculated for model substrate **1J′** with **N1**.

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This rationale is supported by experimental and computational evidence.

The **N4** quenches ^{3}**PC5** much more efficiently than **1a** (K_{SV} = 5120 vs 346 mol^{-1}, respectively, Figure 2c).[27] The lifetime (τ) of ^{3}**PC5** in an 8:2 toluene/DCM solution at room temperature is 1.8 μs. Therefore, the **PC5**-**1a** bimolecular quenching constant k_q is significantly smaller than the **PC5**-**N4** one (2.2×10^8 vs 3.2×10^9 L mol^{-1} s^{-1}, respectively). These results suggest that the initial EnT occurs between the ^{3}**PC5** and the cocatalyst **N4**. According to the DFT modeling data, this step is exergonic by 5.3 kcal mol^{-1}. The same trend was observed quenching the luminescence of ^{3}**PC6**, for which we experimentally measured τ = 0.75 μs in an 8:2 toluene/DCM solution at room temperature. The **PC6**-**1a** bimolecular quenching constant k_q is more than 1 order of magnitude smaller than the **PC6**-**N4** one (3.9×10^8 vs 5.0×10^9 L mol^{-1} s^{-1}, respectively). Analogous quenching studies were performed on both **PC5** and **PC6** to compare the prowess of **N1** and **1A**, which are used in the synthesis of β-lactams (spectra and Stern−Volmer plots in the SI). Once more, the **PC**-**N1** bimolecular quenching constants (2.8×10^9 and 5.0×10^9 L mol^{-1} s^{-1}, for **PC5** and **PC6**, respectively) were more than 1 order of magnitude higher than the corresponding **PC**-**1A** ones (2.4×10^8 and 2.0×10^7 L mol^{-1} s^{-1}, respectively). The results from these eight different sets of luminescence quenching experiments are coherent, and they greatly strengthen the case for an initial EnT between the excited photosensitizer and the **N** cocatalyst, rather than the allenyl substrate, in present reactions.

The ^{3}**N4** and **1** can undergo a second EnT, delivering a sensitized substrate. The phosphorescence of ^{3}**N4** was detected in frozen 2-MeTHF at 77 K, and we measured an E_T (0−0 band) of 56.1 kcal mol^{-1}, which is close to the calculated value of 57.0 kcal mol^{-1}. The result confirms the feasibility of the ^{3}**N4−1** EnT, which is exergonic (the calculated E_T of substrates **1j** and **1J′** was 40.9 kcal mol^{-1} according to DFT data acquired at the M06/Def2-TZVP level, using toluene as implicit solvent and taking into account dispersion interactions via G3 corrections). At 77 K, ^{3}**N4** has a very long lifetime (τ = 1.5 s), suggesting that it is a powerful triplet-state reservoir.[28] It was not possible to directly measure the ^{3}**N4−1a** EnT via Stern−Volmer titration because we were unable to observe the radiative decay of ^{3}**N4** at room temperature (RT), regardless of the solution medium. At RT, **N4** has two fluorescence bands, with maxima at around 330 and 410 nm (Figure 2d). The lifetimes at RT of these two emitting species were 6.7 and 1.8 ns, respectively. The former luminescence band is typical of isolated 1-naphthyl groups, while the latter band has been ascribed to the emission of an excimer of **N4** in which the two aryl groups are interacting.[29] Gratifyingly, the intensity of this band was linearly affected by the addition of **1a** (K_{SV} = 106 mol^{-1}). Static luminescence quenching between the chromophore and the substrate was ruled out because we did not observe shifts in the corresponding absorption and emission UV−vis spectra (spectra in SI). Therefore, the linear correlation between the intensity of the excimer luminescence and the concentration of **1a** indicates that dynamic quenching occurs. This, in turn, implies that the rate of decay of the excimer is slower than the ^{3}**N4−1a** EnT.[30] By taking into account the measured lifetime of the **N4** excimer (τ = 1.8 ns), the bimolecular ^{3}**N4**-**1a** quenching constant k_q is higher than 1.6×10^{10} L mol^{-1} s^{-1}.

The k_q values calculated for the ^{3}**PC5**-**N4**, the ^{3}**PC6**-**N4**, the ^{3}**PC5**-**N1**, the ^{3}**PC6**-**N1**, and the ^{3}**N4−1a** EnTs are all close to diffusion (10^9−10^{10} L mol^{-1} s^{-1}), which is most often the ceiling rate of homogeneous chemical events.[31] These results strongly suggest that present reactions involve a first EnT between the **PC** and the **N** cocatalyst, followed by a second one between the latter and the substrate **1** (Figure 2f summarizes the kinetic scenario involving **PC5**, **N4**, and **1a**). The scenario is consistent with the limited impact of traces of dioxygen on these reactions, especially those delivering β-lactam **3** (Figure 1a). The gas can quench photoexcited PCs at nearly diffusional rates[20] and k_q in the 10^9 L mol^{-1} s^{-1} range were reported (higher for homo—than for the corresponding heteroleptic Ir(III) complexes).[20b] This fast bimolecular reaction hinders any slower EnT event, but the issue becomes less relevant if dioxygen has to compete with quenchers that react at nearly diffusional rates, too.

The relay ensures the efficient activation of challenging substrates, such as terminal allenamides,[12] by replacing a slower EnT (Figures 1a and 2a)[27] with two faster ones. The scenario reminds of the concept of polarity reversal catalysis in HAT reactions,[32] in which a hydrogen abstraction step that has a poor rate is offset by two kinetically fast ones. Although extended aromatics have been used as triplet-state reservoirs,[33] we are unaware of reports proving that the cooperation of two distinct EnT cycles increases the kinetics of substrate sensitization.

The ^{3}**N4−1** EnT can occur through an encounter complex between ^{3}**N4** and the substrate, taking advantage of the long lifetime of the former. Alternatively, a supramolecular adduct, such as [**1j**:**N4**], can form at the ground state and then engage into two sequential EnTs; the first one is an intermolecular event, while the second becomes an intramolecular one, bypassing concentration issues. Both cases led to the formation of the key intermediate of present sequences. According to DFT modeling data (Figure 2e), the sensitization of the adduct [**1j**:**N4**] seems more likely because the latter is more stable than the corresponding separate species **1j** and **N4** by 4.1 kcal mol^{-1} in ΔG. The complexation was confirmed by NMR titration (spectra in SI).[34]

In addition to their role in the substrate activation, the **N** cocatalysts allow the formation of products through energetically favored pathways (blue vs black paths for the complexed vs isolated manifolds, respectively, Figure 2e). For **2**, the transition state is 4.3 kcal mol^{-1} more stable, and ^{3}[**Ij**:**N4**] lies 6.3 kcal mol^{-1} below the separate species ^{3}**Ij** and **N4**. These effects are due to the formation of dispersion interactions between the π clouds of **N4** and the mono-occupied molecular orbitals (green basins in the Non Covalent Interactions plots[35]). Similarly, the formation of β-lactam **3J′** involves a lower barrier, by −1.1 kcal mol^{-1} in ΔG, for the [**1J′**:**N1**] adduct, while the intermediates are slightly endergonic.

The role of dispersion interactions between mono-occupied orbitals and the π-clouds of naphthyl groups has been proposed by recent DFT studies on the photoisomerization of styrenes,[13a] alkene−alkene [2 + 2][13b] and dearomative [4 + 2] cycloadditions.[12c] In the present case, the formation of adducts with the **N** cocatalyst finds an additional experimental support. The variation of the **2**:**3** ratio obtained switching **N1** with **N4** (Figure 1c) is indicative of an interaction between the cocatalyst and the transient triplets of the sequence, such as in the blue pathways of Figure 2e; the variation of the nature of the $\mathbf{N_n}$ affects the relative convenience of two competing **TS**(**1**-

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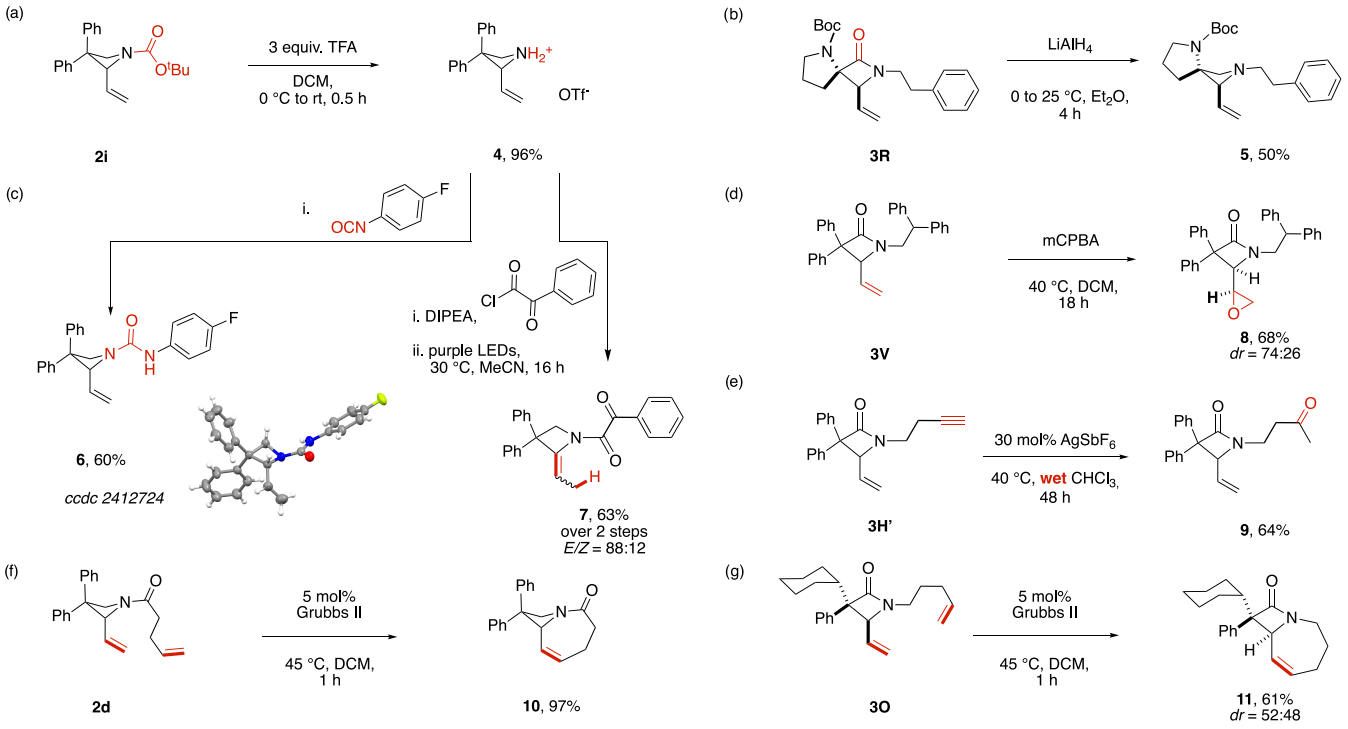


Figure 3. Representative model derivatizations of vinyl azetidines and β-lactams.

I:N$_n$), and thus alters the ratio of the two products. On the contrary, the same **2**:**3** ratio would have been observed using either **N1** or **N4** if the **N$_n$** did not interact with the intermediates of the sequence (black pathways in Figure 2e) because the **2**:**3** ratio would be dictated by the relative convenience of the two competing **TS(1-I)**, which are the only transition state of the two orthogonal reactions (Figure 1c, red vs green HATs).

The derivatization of α-vinyl azetidines[4] **2** and β-vinyl β-lactams **3** was used to extend the accessible chemical space (Figure 3).

The carbamate group of **2i** can be easily deprotected, delivering NH-free azetidine **4** with an excellent yield. The amide reduction on **3R** furnished spirocyclic azetidine **5**. The azetidine **4** has many applications (Scheme 1b),[4] and it can be derivatized with electrophiles. For instance, the use of an isocyanate gives urea **6**; treatment with phenylglyoxyl chloride allows us to trigger a visible-light-promoted[36] formal allylic isomerization (**7**). This provided a valuable[6] vinylidene-azetidine with two contiguous quaternary carbons. The vinyl group can be treated with a peracid, delivering epoxide **8** with good diastereocontrol. A mild alkyne hydration[37] leaves the heterocyclic core unscathed (**9**). Finally, fused bicyclic products **10** and **11**, featuring a prized headbridging nitrogen nucleus,[1b,e] could be smoothly prepared via ring-closing metathesis.

CONCLUSIONS

We presented a simple visible-light-mediated preparation of densely functionalized azetidines and β-lactams, which present a valuable allylamine unit, from allenamides. A dual EnT process allows one to synthesize these challenging heterocycles at room temperature with broad functional group tolerance. The original HAT sequence involves a vinyl-radical site, which has been generated in an atom-economical fashion. The method combines a powerful Ir(III) photosensitizer, which was not previously exploited in EnT reactions, with a binaphthyl cocatalyst, which could stabilize the intermediates of the sequence with its π clouds. Moreover, it ensures efficient substrate activation by relaying two EnT cycles. We anticipate the broad application of the relay concept for the synthesis of elaborate molecular architectures.

METHODS

General Procedure for the Synthesis of Azetidines 2. A vial was charged with the desired allene **1** (0.15 mmol, 1 equiv), the **PC5** (1.5 mg, 1 mol %), and the cocatalyst **N4** (8.9 mg, 30 mol %). A toluene/DCM 8:2 solution was then added (1.5 mL, 0.1 M on substrate), and the resulting solution was transferred via syringe into an NMR tube capped with a rubber septum. The mixture was degassed by freeze−pump−thaw (3 times) using a needle connected to a Schlenk line to pierce the septum and then irradiated with purple LEDs until complete consumption of the starting material, as monitored by TLC (14−72 h). The solution was then transferred into a round-bottomed flask via syringe, the NMR tube was rinsed with DCM (4 × 1 mL), and the resulting mixture was concentrated under reduced pressure. The crude product was purified via column chromatography on silica gel (*n*-hexane/EtOAc, under gradient) to afford the desired product **2**.

General Procedure for the Synthesis of β-Lactams 3. A vial was charged with the desired allene **1** (0.15 mmol, 1 equiv), the **PC5** (1.5 mg, 1 mol %), and the cocatalyst **N1** (8.5 mg, 30 mol %). A toluene/DCM 8:2 solution was then added (1.5 mL, 0.1 M on substrate), and the resulting solution was transferred via syringe into an NMR tube capped with a rubber septum. The mixture was briefly flushed with N_2 using a needle connected to a Schlenk line to pierce the septum and then irradiated with purple LEDs until complete consumption of the starting material, as monitored by TLC (24−72 h). The

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solution was then transferred into a round-bottomed flask via syringe, the NMR tube was rinsed with DCM (4 × 1 mL), and the resulting mixture was concentrated under reduced pressure. The crude product was purified via column chromatography on silica gel (*n*-hexane/EtOAc, under gradient) to afford the desired product **3**.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c04424.

Detailed optimization studies, synthesis of substrates, characterization of products, luminescence studies, computational data, copies of NMR spectra (PDF)

Prod **2i** (CIF)

Prod **3A** (CIF)

Prod **6** (CIF)

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†S.P. and G.S. contributed equally to this work. G.M. conceived the project with contributions from all authors; S.S. and G.S., with contribution from A.C. and G.R., performed synthetic experiments; D.S., with contribution from S.S. and G.R., performed spectroscopic studies; L.M. performed crystallographic analyses; M.L. and G.M. performed computational studies; and G.M. wrote the manuscript with contributions from all authors. All authors reviewed the data.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) For selected examples involving azetidines, see (a) Lescop, C.; Birker, M.; Brotschi, C.; Bürki, C.; Morrison, K.; Froidevaux, S.; Delahaye, S.; Nayler, O.; Bolli, M. H. Discovery of the Novel, Orally Active, and Selective LPA1 Receptor Antagonist ACT-1016−0707 as a Preclinical Candidate for the Treatment of Fibrotic Diseases. *J. Med. Chem.* **2024**, *67*, 2397−2424. (b) Maetani, M.; Zoller, J.; Melillo, B.; Verho, O.; Kato, N.; Pu, J.; Comer, E.; Schreiber, S. L. Synthesis of a Bicyclic Azetidine with In Vivo Antimalarial Activity Enabled by Stereospecific, Directed C(sp^3)−H Arylation. *J. Am. Chem. Soc.* **2017**, *139*, 11300−11306. For b-lactams, see (c) Mora-Ochomogo, M.; Lohans, C. T. *β*-Lactam antibiotic targets and resistance mechanisms: from covalent inhibitors to substrates. *RSC Med. Chem.* **2021**, *12*, 1623−1639. For reviews, see (d) Brandi, A.; Cicchi, S.; Cordero, F. M. Novel Syntheses of Azetidines and Azetidinones. *Chem. Rev.* **2008**, *108*, 3988−4035. (e) Pitts, C. R.; Lectka, T. Chemical Synthesis of *β*-Lactams: Asymmetric Catalysis and Other Recent Advances. *Chem. Rev.* **2014**, *114*, 7930−7953. (f) Hosseyni, S.; Jarrahpour, A. Recent advances in *β*-lactam synthesis. *Org. Biomol. Chem.* **2018**, *16*, 6840−6852. (g) Richardson, A. D.; Becker, M. R.; Schindler, C. Synthesis of azetidines by aza Paternò−Büchi reactions. *Chem. Sci.* **2020**, *11*, 7553−7561. (h) Lima, L. M.; da Silva, B. N. M.; Barbosa, G.; Barreiro, E. J. *β*-lactam antibiotics: An overview from a medicinal chemistry perspective. *Eur. J. Med. Chem.* **2020**, *208*, No. 112829. (i) Mughal, H.; Szostak, M. Recent advances in the synthesis and reactivity of azetidines: strain-driven character of the four-membered heterocycle. *Org. Biomol. Chem.* **2021**, *19*, 3274−3286.

(2) (a) Stütz, A. Allylamine Derivatives—a New Class of Active Substances in Antifungal Chemotherapy. *Angew. Chem., Int. Ed.* **1987**, *26*, 320−328. (b) Johannsen, M.; Jørgensen, K. A. Allylic Amination. *Chem. Rev.* **1998**, *98*, 1689−1708. (c) Candeias, N. R.; Montalbano, F.; Cal, P. M. S. D.; Gois, P. M. P. Boronic Acids and Esters in the Petasis-Borono Mannich Multicomponent Reaction. *Chem. Rev.* **2010**, *110*, 6169−6193. (d) Nag, S.; Batra, S. Applications of allylamines for the syntheses of aza-heterocycles. *Tetrahedron* **2011**, *67*, 8959−9061. (e) Skoda, E. M.; Davis, G. C.; Wipf, P. Allylic Amines as Key Building Blocks in the Synthesis of (E)-Alkene Peptide Isosteres. *Org. Process Res. Dev.* **2012**, *16*, 26−34.

(3) For representative linear approaches to azetidines, see (a) Ju, Y.; Varma, R. S. Aqueous N-heterocyclization of Primary Amines and Hydrazines with Dihalides: Microwave-assisted Syntheses of N-Azacycloalkanes, Isoindole, Pyrazole, Pyrazolidine, and Phthalazine Derivatives. *J. Org. Chem.* **2006**, *71*, 135−141. (b) Fritz, S. P.; Moya, J. F.; Unthank, M. G.; McGarrigle, E. M.; Aggarwal, V. K. An Efficient Synthesis of Azetidines with (2-Bromoethyl)sulfonium Triflate. *Synthesis* **2012**, *44*, 1584−1590. (c) Betz, K. N.; Chiappini, N. D.; Du Bois, J. Intermolecular sp3-C−H Amination for the Synthesis of Saturated Azacycles. *Org. Lett.* **2020**, *22*, 1687−1691. (d) Xu, P.; Zhang, M.; Ingoglia, B.; Allais, C.; Dechert-Schmitt, A.-M. R.; Singer, R. A.; Morken, J. P. Construction of Azacycles by Intramolecular Amination of Organoboronates and Organobis(boronates). *Org. Lett.* **2021**, *23*, 3379−3383. For the synthesis of a-vinyl azetidines, see (e) Dolle, R. E.; Li, C.-S.; Shaw, A. N. Concomitant [2,3]-sigmatropic rearrangement of allylic sulfilimines and intramolecular N-alkylation. Synthesis of 2-vinyl substituted cyclic amines. *Tetrahedron Lett.* **1989**,

30, 4723−4726. (f) Hassner, A.; Weigand, N. Small rings. 29. Synthesis and ring expansion of vinylazetidines. A synthesis of hydroazocines. *J. Org. Chem.* **1986**, *51*, 3652−3656.
(4) With metal-carbene complexes, see (a) Rivero, A. R.; Fernandez, I.; Sierra, M. A. The Photochemical Reaction of Vinylaziridines and Vinylazetidines with Chromium(0) and Molybdenum(0) (Fischer) Carbene Complexes. *Chem. - Eur. J.* **2014**, *20*, 1359−1366. With arynes, see (b) Aoki, T.; Koya, S.; Yamasaki, R.; Saito, S. Cycloaddition Reaction of 2-Vinylazetidines with Benzyne: A Facile Access to 1-Benzazocine Derivatives. *Org. Lett.* **2012**, *14*, 4506−4509. With strong bases from azetidiniums, see (c) Couty, F.; Durrat, F.; Evano, G.; Marrot, J. Ring Expansions of 2-Alkenylazetidinium Salts − a New Route to Pyrrolidines and Azepanes. *Eur. J. Org. Chem.* **2006**, *2006*, 4214−4223. For Pd-catalysis, see (d) Satake, A.; Ishii, H.; Shimizu, I.; Inoue, Y.; Hasegawa, H.; Yamamoto, A. Palladium-catalyzed ring-opening reactions of 1-acetyl-4-vinyl-2-azetidinones and 1-sulfonyl-2-vinylazetidines. Role of intramolecular participation of amide anion. *Tetrahedron* **1995**, *51*, 5331−5340.
(5) For recent nonphotocatalytic approaches using the strain release concept, see (a) Ning, Y.; Chen, H.; Ning, Y.; Zhang, J.; Bi, X. Rhodium-Catalyzed One-Carbon Ring Expansion of Aziridines with Vinyl-N-triftosylhydrazones for the Synthesis of 2-Vinyl Azetidines. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202318072. (b) Hsu, C.-M.; Lin, H.-B.; Hou, X.-Z.; Tapales, R. V. P. P.; Shih, C.-K.; Miñoza, S.; Tsai, Y.-S.; Tsai, Z.-N.; Chan, C.-L.; Liao, H.-H. Azetidines with All-Carbon Quaternary Centers: Merging Relay Catalysis with Strain Release Functionalization. *J. Am. Chem. Soc.* **2023**, *145*, 19049−19059. (c) Jaiswal, V.; Mondal, S.; Singh, B.; Singh, V. P.; Saha, J. Cation-Promoted Strain-Release-Driven Access to Functionalized Azetidines from Azabicyclo[1.1.0]butanes. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202304471. (d) Tyler, J. L.; Noble, A.; Aggarwal, V. K. Strain-Release Driven Spirocyclization of Azabicyclo[1.1.0]butyl Ketones. *Angew. Chem., Int. Ed.* **2021**, *60*, 11824−11829. (e) Fawcett, A.; Murtaza, A.; Gregson, C. H. U.; Aggarwal, V. K. Strain-Release-Driven Homologation of Boronic Esters: Application to the Modular Synthesis of Azetidines. *J. Am. Chem. Soc.* **2019**, *141*, 4573−4578.
(6) Jacob, C.; Baguia, H.; Dubart, A.; Oger, S.; Thilmany, P.; Beaudelot, J.; Deldaele, C.; Peruško, S.; Landrain, Y.; Michelet, B.; Neale, N.; Romero, E.; Moucheron, C.; Van Speybroeck, V.; Theunissen, C.; Evano, G. A general synthesis of azetidines by copper-catalysed photoinduced anti-Baldwin radical cyclization of ynamides. *Nat. Commun.* **2022**, *13*, No. 560.
(7) For recent reviews on EnT, see (a) Dutta, S.; Erchinger, J. E.; Strieth-Kalthoff, F.; Kleinmans, R.; Glorius, F. Energy transfer photocatalysis: exciting modes of reactivity. *Chem. Soc. Rev.* **2024**, *53*, 1068−1089. (b) Yadav, R. N.; Srivastava, A. K.; Dey, S.; Das, A.; Banik, B. K.; Hossain, F. Transition metal complexes: A new era of photosensitizers for Dearomative photocycloaddition/annulation via energy and Electron transfer Photocatalysis. *Coord. Chem. Rev.* **2024**, *520*, No. 216136. (c) Großkopf, J.; Kratz, T.; Rigotti, T.; Bach, T. Enantioselective Photochemical Reactions Enabled by Triplet Energy Transfer. *Chem. Rev.* **2022**, *122*, 1626−1653. (d) Neveselý, T.; Wienhold, M.; Molloy, J. J.; Gilmour, R. Advances in the E → Z Isomerization of Alkenes Using Small Molecule Photocatalysts. *Chem. Rev.* **2022**, *122*, 2650−2694. (e) Strieth-Kalthoff, F.; Glorius, F. Triplet Energy Transfer Photocatalysis: Unlocking the Next Level. *Chem* **2020**, *6*, 1888−1903. (f) Zhou, Q.-Q.; Zou, Y.-Q.; Lu, L.-Q.; Xiao, W.-J. Visible-Light-Induced Organic Photochemical Reactions through Energy-Transfer Pathways. *Angew. Chem., Int. Ed.* **2019**, *58*, 1586−1604.
(8) Rodríguez, R. I.; Corti, V.; Rizzo, L.; Visentini, S.; Bortolus, M.; Amati, A.; Natali, M.; Pelosi, G.; Costa, P.; Dell'Amico, L. Radical strain-release photocatalysis for the synthesis of azetidines. *Nat. Catalysis* **2024**, *7*, 1223−1231.
(9) Wearing, E. R.; Yeh, Y.-C.; Terrones, G. G.; Parikh, S. G.; Kevlishvili, I.; Kulik, H. J.; Schindler, C. S. Visible light-mediated aza Paternò-Büchi reaction of acyclic oximes and alkenes to azetidines. *Science* **2024**, *384*, 1468−1476. For an intramolecular approach activating styryl arms, see (b) Gu, X.; Shen, J.; Xu, Z.; Liu, J.; Shi, M.; Wei, Y. Visible-Light-Mediated Activation of Remote C(sp3)−H Bonds by Carbon-Centered Biradical via Intramolecular 1,5- or 1,6-Hydrogen Atom Transfer. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202409463.
(10) For reviews on photochemical HATs, see (a) Capaldo, L.; Ravelli, D.; Fagnoni, M. Direct Photocatalyzed Hydrogen Atom Transfer (HAT) for Aliphatic C−H Bonds Elaboration. *Chem. Rev.* **2022**, *122*, 1875−1924. (b) Cao, H.; Tang, X.; Tang, H.; Yuan, Y.; Wu, J. Photoinduced intermolecular hydrogen atom transfer reactions in organic synthesis. *Chem. Catal.* **2021**, *1*, 523−598. (c) Sarkar, S.; Cheung, K. P. S.; Gevorgyan, V. C−H functionalization reactions enabled by hydrogen atom transfer to carbon-centered radicals. *Chem. Sci.* **2020**, *11*, 12974−12993.
(11) (a) Oddy, M. J.; Kusza, D. A.; Epton, R. G.; Lynam, J. M.; Unsworth, W. P.; Petersen, W. F. Visible-Light-Mediated Energy Transfer Enables the Synthesis of β-Lactams via Intramolecular Hydrogen Atom Transfer. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202213086. (b) Li, H.-Y.; Zhang, S.-Z.; Niu, X.; Meng, Q.-Y.; Yang, X.-L. Visible Light Photocatalyzed Norrish Type II Reaction of Acrylamides for the Synthesis of β-Lactams. *Adv. Synth. Catal.* **2025**, *367*, No. e202400976.
(12) For EnT on C-allenyl amides, see (a) Plaza, M.; Großkopf, J.; Breitenlechner, S.; Bannwarth, C.; Bach, T. Photochemical Deracemization of Primary Allene Amides by Triplet Energy Transfer: A Combined Synthetic and Theoretical Study. *J. Am. Chem. Soc.* **2021**, *143*, 11209−11217. For EnT on acyl allenamides, see (b) Serafino, A.; Chiminelli, M.; Balestri, D.; Marchiò, L.; Bigi, F.; Maggi, R.; Malacria, M.; Maestri, G. Dimerizing cascades of enallenamides reveal the visible-light-promoted activation of cumulated C−C double bonds. *Chem. Sci.* **2022**, *13*, 2632−2639. (c) Chiminelli, M.; Serafino, A.; Ruggeri, D.; Marchiò, L.; Bigi, F.; Maggi, R.; Malacria, M.; Maestri, G. Visible-Light Promoted Intramolecular para-Cyclo-additions on Simple Aromatics. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202216817. For EnT on sulfonyl allenamides, see (d) Guo, J.-D.; Korsaye, F.-A.; Schutz, D.; Ciofini, I.; Miesch, L. Photocatalyst-free, visible-light-induced regio- and stereoselective synthesis of phosphorylated enamines from N-allenamides via [1,3]-sulfonyl shift at room temperature. *Chem. Sci.* **2024**, *15*, 17962−17970.
(13) (a) Cerveri, A.; Scarica, G.; Sparascio, S.; Hoch, M.; Chiminelli, M.; Tegoni, M.; Protti, S.; Maestri, G. Boosting Energy-Transfer Processes via Dispersion Interactions. *Chem. - Eur. J.* **2024**, *30*, No. e202304010. (b) Hoch, M.; Sparascio, S.; Cerveri, A.; Bigi, F.; Maggi, R.; Viscardi, R.; Maestri, G. The effect of tethered bi-naphthyls on visible-light promoted alkene-alkene [2 + 2] cycloadditions. *Photochem. Photobiol. Sci.* **2024**, *23*, 1543−1563.
(14) (a) Blanksby, S. J.; Ellison, G. B. Bond Dissociation Energies of Organic Molecules. *Acc. Chem. Res.* **2003**, *36*, 255−263. An online practical tool for the calculation of BDE and BD free energies, based on the article (b) St John, P. C.; Guan, Y.; Kim, Y.; Kim, S.; Paton, R. S. Prediction of organic homolytic bond dissociation enthalpies at near chemical accuracy with sub-second computational cost. *Nat. Commun.* **2020**, *11*, No. 2328.
(15) Fore recent examples, see (a) Li, B.; Xu, H.; Dang, Y.; Houk, K. N. Dispersion and Steric Effects on Enantio-/Diastereoselectivities in Synergistic Dual Transition-Metal Catalysis. *J. Am. Chem. Soc.* **2022**, *144*, 1971−1985. (b) Xu, E. Y.; Werth, J.; Roos, C. B.; Bendelsmith, A. J.; Sigman, M. S.; Knowles, R. R. Noncovalent Stabilization of Radical Intermediates in the Enantioselective Hydroamination of Alkenes with Sulfonamides. *J. Am. Chem. Soc.* **2022**, *144*, 18948−18958. (c) Eschmann, C.; Song, L.; Schreiner, P. R. London Dispersion Interactions Rather than Steric Hindrance Determine the Enantioselectivity of the Corey−Bakshi−Shibata Reduction. *Angew. Chem., Int. Ed.* **2021**, *60*, 4823−4832. (d) Pölloth, B.; Sibi, M. P.; Zipse, H. The Size-Accelerated Kinetic Resolution of Secondary Alcohols. *Angew. Chem., Int. Ed.* **2021**, *60*, 774−778. For a review, see (e) Rummel, L.; Schreiner, P. R. Advances and Prospects in Understanding London Dispersion Interactions in Molecular Chemistry. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202316364.

https://doi.org/10.1021/acscatal.5c04424
ACS Catal. 2025, 15, 13799−13809

(16) Mateos, J.; Rigodanza, F.; Costa, P.; Natali, M.; Vega-Peñaloza, A.; Fresch, E.; Collini, E.; Bonchio, M.; Sartorel, A.; Dell'Amico, L. Unveiling the impact of the light source and steric factors on [2 + 2] heterocycloaddition reactions. *Nat. Synth.* **2023**, *2*, 23−36.
(17) (a) Wang, W.; Cai, Y.; Guo, R.; Brown, M. K. Synthesis of complex bicyclic scaffolds by intermolecular photosensitized dearomative cycloadditions of activated alkenes and naphthalenes. *Chem. Sci.* **2022**, *13*, 13582−13587. (b) Guo, R.; Chang, Y.-C.; Herter, L.; Salome, C.; Braley, S. E.; Fessard, T. C.; Brown, M. K. Strain-Release [2π + 2σ] Cycloadditions for the Synthesis of Bicyclo[2.1.1]hexanes Initiated by Energy Transfer. *J. Am. Chem. Soc.* **2022**, *144*, 7988−7994. (c) Chang, Y.-C.; Salome, C.; Fessard, T. C.; Brown, M. K. Synthesis of 2-Azanorbornanes via Strain-Release Formal Cycloadditions Initiated by Energy Transfer. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202314700.
(18) Coppo, P.; Plummer, E. A.; De Cola, L. Tuning iridium(iii) phenylpyridine complexes in the "almost blue" region. *Chem. Commun.* **2004**, 1774−1775.
(19) For PC5, see (a) Di Luzio, S.; Mdluli, V.; Connell, T. U.; Lewis, J.; VanBenschoten, V.; Bernhard, S. High-Throughput Screening and Automated Data-Driven Analysis of the Triplet Photophysical Properties of Structurally Diverse, Heteroleptic Iridium(III) Complexes. *J. Am. Chem. Soc.* **2021**, *143*, 1179−1194. For PC6, see (b) Tamayo, A. B.; Alleyne, B. D.; Djurovich, P. I.; Lamansky, S.; Tsyba, I.; Ho, N. N.; Bau, R.; Thompson, M. E. Synthesis and Characterization of Facial and Meridional Tris-cyclometalated Iridium(III) Complexes. *J. Am. Chem. Soc.* **2003**, *125*, 7377−7387. (c) We warmly thank a reviewer for the useful suggestion to test complex PC6, for which an E_T = 63.5 kcal/mol was reported, see Zhu, M.; Zhang, X.; Zheng, C.; You, S.-L. Visible-Light-Induced Dearomatization via [2 + 2] Cycloaddition or 1,5-Hydrogen Atom Transfer: Divergent Reaction Pathways of Transient Diradicals. *ACS Catal.* **2020**, *10*, 12618−12626. (d) For PC7, see Graf, M.; Czerwieniec, R.; Mayer, P.; Bottcher, H.-C. Synthesis and structural characterization of bis-cyclometalated compounds [Ir(dFppy)$_2$(Me$_4$phen)]PF$_6$ and [Ir(dF(CF$_3$)ppy)$_2$(Me$_4$phen)]PF$_6$. *Inorg. Chim. Acta* **2021**, *527*, No. 120554.
(20) (a) Ashen-Garry, D.; Selke, M. Singlet Oxygen Generation by Cyclometalated Complexes and Applications. *Photochem. Photobiol.* **2014**, *90*, 257−274. (b) Hasebe, N.; Deguchi, Y.; Murayama, S.; Yoshihara, T.; Horiuchi, H.; Okutsu, T.; Tobita, S. Phosphorescence quenching of neutral and cationic iridium(III) complexes by molecular oxygen and aromatic electron acceptors. *J. Photochem. Photobio. A Chem.* **2016**, *324*, 131−144.
(21) CCDC Deposition numbers 2412725 (for 2i), 2412723 (for 3a), and 2412724 (for 6) contain the supplementary crystallographic data for this paper. These data are provided free of charge by the Cambridge Crystallographic Data Centre.
(22) (a) Dampalla, C. S.; Rathnayake, A. D.; Kankanamalage, A. C. G.; Kim, Y.; Perera, K. D.; Nguyen, H. N.; Miller, M. J.; Madden, T. K.; Picard, H. R.; Truman, H. A.; Kashapathy, M. M.; Liu, L.; Battaile, K. P.; Lovell, S.; Chang, K.-O. Structure-Guided Design of Potent Spirocyclic Inhibitors of Severe Acute Respiratory Syndrome Coronavirus-2 3C-like Protease. *J. Med. Chem.* **2022**, *65*, 7818−7832. (b) Kirichok, A. A.; Tkachuk, H.; Kozyriev, Y.; Shablykin, O.; Datsenko, O.; Granat, D.; Yegorova, T.; Bas, Y. P.; Semirenko, V.; Pishel, I.; Kubyshkin, V.; Lesyk, D.; Klymenko-Ulianov, O.; Mykhailiuk, P. K. 1-Azaspiro[3.3]heptane as a Bioisostere of Piperidine. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202311583.
(23) (a) Miyashita, M.; Sasaki, M.; Hattori, I.; Sakai, M.; Tanino, K. Total synthesis of norzoanthamine. *Science* **2004**, *305*, 495−499. (b) Wood, M. E.; Bissiriou, S.; Lowe, C.; Norrish, A. M.; Sénéchal, K.; Windeatt, K. M.; Coles, S. J.; Hursthouse, M. B. Synthetic use of the primary kinetic isotope effect in hydrogen atom transfer: generation of α-aminoalkyl radicals. *Org. Biomol. Chem.* **2010**, *8*, 4653−4665.
(24) (a) DeZutter, C. B.; Horner, J. H.; Newcomb, M. Rate Constants for 1,5- and 1,6-Hydrogen Atom Transfer Reactions of Mono-, Di-, and Tri-aryl-substituted Donors, Models for Hydrogen Atom Transfers in Polyunsaturated Fatty Acid Radicals. *J. Phys. Chem. A* **2008**, *112*, 1891−1896. (b) Simmons, E. M.; Hartwig, J. F. On the Interpretation of Deuterium Kinetic Isotope Effects in C-H Bond Functionalizations by Transition-Metal Complexes. *Angew. Chem., Int. Ed.* **2012**, *51*, 3066−3072.
(25) Dispersion corrections were applied according to the method by Grimme, see (a) Grimme, S. *WIREs Comput. Mol. Sci.* **2011**, *1*, 211−218. doi: Density functional theory with London dispersion corrections (b) Grimme, S.; Ehrlich, S.; Goerigk, L. Journal of Computational ChemistryEffect of the damping function in dispersion corrected density functional theory. *J. Comput. Chem.* **2011**, *32*, 1456−1465.
(26) For selected studies on $\pi-\pi$ stacking and C H-π interactions, see (a) Bravin, C.; Piekos, J. A.; Licini, G.; Hunter, C. A.; Zonta, C. Dissection of the Polar and Non-Polar Contributions to Aromatic Stacking Interactions in Solution. *Angew. Chem., Int. Ed.* **2021**, *60*, 23871−23877. (b) Fatima, M.; Steber, A. L.; Poblotzki, A.; Pérez, C.; Zinn, S.; Schnell, M. Rotational Signatures of Dispersive Stacking in the Formation of Aromatic Dimers. *Angew. Chem., Int. Ed.* **2019**, *58*, 3108−3145. (c) Hwang, J.; Dial, B. E.; Li, P.; Kozik, M. E.; Smith, M. D.; Shimizu, K. D. How important are dispersion interactions to the strength of aromatic stacking interactions in solution? *Chem. Sci.* **2015**, *6*, 4358−4364.
(27) In previous studies on the EnT-activation of allenamides in the presence of naphthalene, and β-styrenes in the presence of naphthalene or poly-naphthyl N derivatives, refs 12c and 13, organic substrates were better quenchers than N for the excited Ir(ppy)$_3$ (PC1) photocatalyst; because ^{3}PC1 is not quenched by N4 while ^{3}PC5 is, the comparison of entries in Tables S1a and S1e in the SI mostly reflects the impact of radical-π dispersion interactions on the reaction, while the comparison of entry 19 with entry 26 in Table S1f in the SI shows the role of the EnT relay manifold on the activation of 1; the starkly different outcome between the two entries under best conditions (76% vs 27%) is due to the combination of the effect of the N co-catalyst.
(28) We were unable to measure a lifetime for ^{3}N4 at room temperature in various media, but Davidson reported lifetimes of 14.0 and 14.5 μs for ^{3}N4 and ^{3}N1, respectively, at room temperature in cyclohexane; these values are compatible with a role of triplet-state reservoir because they are much longer than the lifetime of excited Ir(III) photosensitizers, see Beecroft, R. A.; Davidson, R. S.; Goodwin, D. The photochemistry of intramolecular excimers: the role of intersystem crossing and product formation. *Tetrahedron* **1984**, *40*, 4497−4500.
(29) (a) Todesco, B.; Gelan, J.; Martens, H.; Put, J.; Boens, N.; De Schryver, F. C. Photochemistry of non conjugated bichromophoric systems photocyclomerization of di-(α-naphthylmethyl)ether. *Tetrahedron Lett.* **1978**, *19*, 2815−2818. For a study on the luminescence properties of N1, see (b) Subudhi, P. C.; Lim, E. C. Conformational difference between singlet and triplet excimers of naphthalene as revealed by luminescence of 1,2-di-α-naphthylethane. *Chem. Phys. Lett.* **1976**, *44*, 479−481.
(30) The emission of the *N4 excimer does not overlap with the absorption 1, suggesting that a Förster-type EnT should not occur; the limited overlap between the emission of the *N4 excimer and the emission of 1 reduces the likeliness that a singlet−singlet EnT might be responsible for the luminesence quenching of *N4 by the substrate; the feasibility of the *Nn-to-^{3}Nn intersystem crossing, which is required prior to a ^{3}N4−1 EnT, is described in refs 28 and 29b; similarly to the approach described herein for N4, the assessment of triplet−triplet EnT via Stern−Volmer titration quenching a fluorescence band, rather than a phosphorescence one, is routinely carried out for thioxanthone, which is a popular photosensitizer; see, for instance (a) Kleinmans, R.; Katzenburg, F.; Daniliuc, C.; Glorius, F.; et al. Energy Transfer-Enabled Unsymmetrical Diamination Using Bifunctional Nitrogen-Radical Precursors. *Nat. Catal.* **2022**, *5*, 1120−1130. (b) Tan, G.; Das, M.; Keum, H.; Belotti, P.; Daniliuc, C.; Glorius, F. Photochemical single-step synthesis of β-amino acid derivatives from alkenes and (hetero)arenes. *Nat. Chem.* **2022**, *14*, 1174−1184.

https://doi.org/10.1021/acscatal.5c04424
ACS Catal. 2025, 15, 13799−13809

(31) Elliot, A. J.; McCraken, D. R.; Buxton, G. V.; Wood, N. D. Estimation of rate constants for near-diffusion-controlled reactions in water at high temperatures. *J. Chem. Soc., Faraday Trans.* **1990**, *86*, 1539−1547.
(32) (a) Roberts, B. P. Polarity-reversal catalysis of hydrogen-atom abstraction reactions: concepts and applications in organic chemistry. *Chem. Soc. Rev.* **1999**, *28*, 25−35. (b) Pan, X.; Lacôte, E.; Lalevée, J.; Curran, D. P. Polarity Reversal Catalysis in Radical Reductions of Halides by N-Heterocyclic Carbene Boranes. *J. Am. Chem. Soc.* **2012**, *134*, 5669−5674.
(33) (a) Parker, C. A.; Bowen, E. J. Sensitized P-type delayed fluorescence. *Proc. R. Soc. A* **1963**, *276*, 125−135. (b) Wang, C.; Reichenauer, F.; Kitzmann, W. R.; Kerzig, C.; Heinze, K.; Resch-Genger, U. Efficient Triplet-Triplet Annihilation Upconversion Sensitized by a Chromium(III) Complex via an Underexplored Energy Transfer Mechanism. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202202238. (c) Li, H.; Wang, C.; Glaser, F.; Sinha, N.; Wenger, O. S. Metal−Organic Bichromophore Lowers the Upconversion Excitation Power Threshold and Promotes UV Photoreactions. *J. Am. Chem. Soc.* **2023**, *145*, 11402−11414.
(34) In contrast to supramolecular adducts featuring strong charge-transfer interactions, such as donor-acceptor complexes, whose complexation equilibria are easily studied by UV−vis titration, the study of equilibria involving adducts driven by relatively weaker $\pi-\pi$ interactions is commonly achieved through lower-energy spectroscopic techniques, such as NMR and rotational spectroscopy. See, for instance, references 26b-c.
(35) Contreras-García, J.; Johnson, E. R.; Keinan, S.; Chaudret, R.; Piquemal, J.-P.; Beratan, D.; Yang, W. NCIPLOT: A Program for Plotting Noncovalent Interaction Regions. *J. Chem. Theory Comput.* **2011**, *7*, 625−632.
(36) For the photoactivation of phenylglyoxylates, see Hu, S.; Neckers, D. C. Photochemical Reactions of Alkyl Phenylglyoxylates. *J. Org. Chem.* **1996**, *61*, 6407−6415.
(37) Reported Ag(I) catalysts for alkyne hydration require warming up to 80−110 °C, see (a) Das, R.; Chakraborty, D. AgOTf catalyzed hydration of terminal alkynes. *Appl. Organomet. Chem.* **2012**, *26*, 722−726. (b) Chen, Z.-W.; Ye, D.-N.; Qian, Y.-P.; Ye, M.; Liu, L.-X. Highly efficient AgBF4-catalyzed synthesis of methyl ketones from terminal alkynes. *Tetrahedron* **2013**, *69*, 6116−6120.



https://doi.org/10.1021/acscatal.5c04424
ACS Catal. 2025, 15, 13799−13809